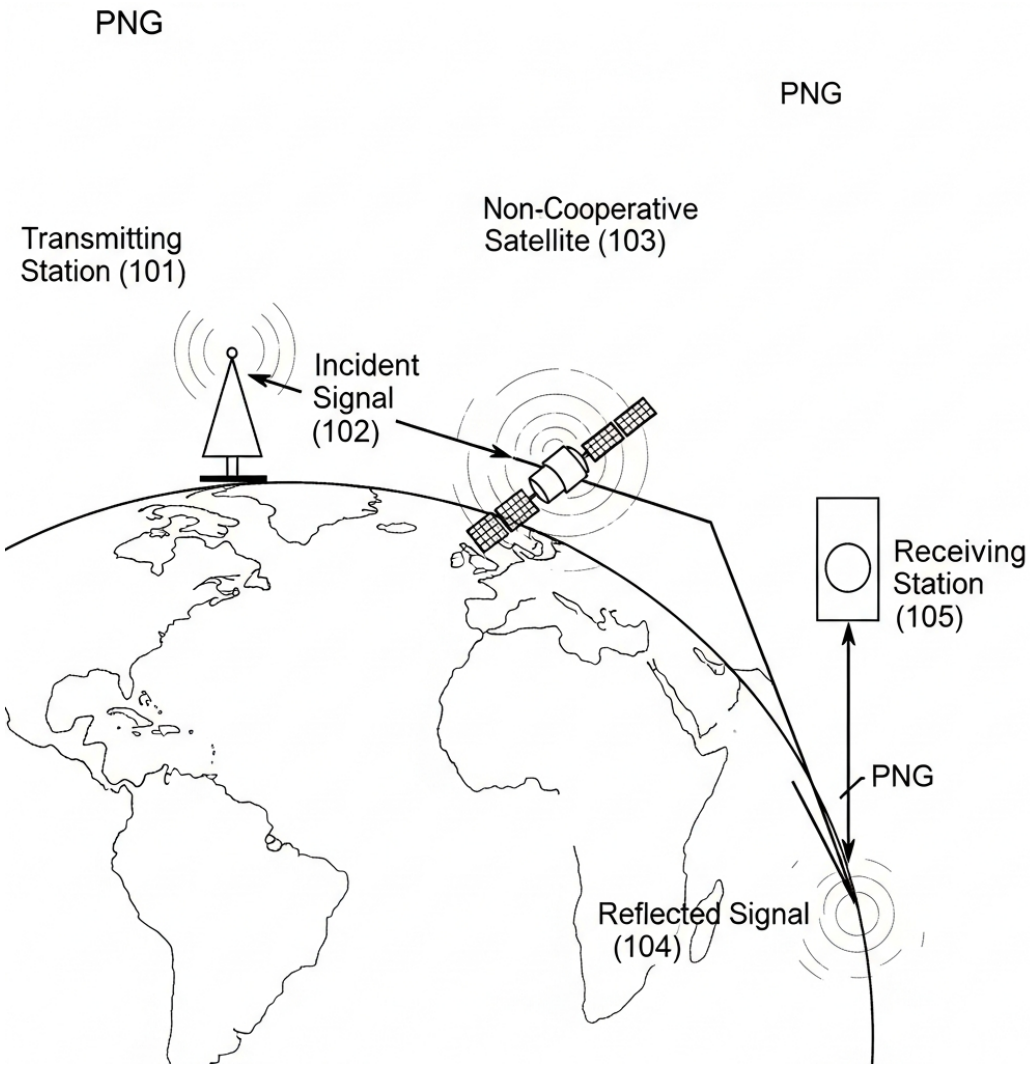

PNG
PNG
Non-Cooperative
Satellite (103)
Transmitting
Station (101)
Incident
Signal
(102)
Receiving
Station
(105)
PNG
Reflected Signal
(104)

FIG. 1

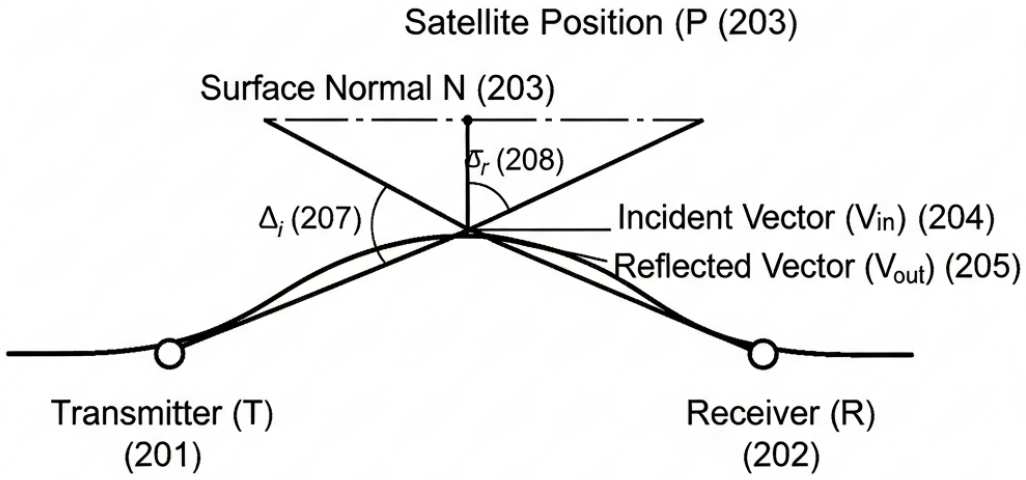

Satellite Position (P (203)
Surface Normal N (203)
σ_r (208)
Δ_i (207)
Incident Vector (V_{in}) (204)
Reflected Vector (V_{out}) (205)
Transmitter (T)
(201)
Receiver (R)
(202)

FIG. 2

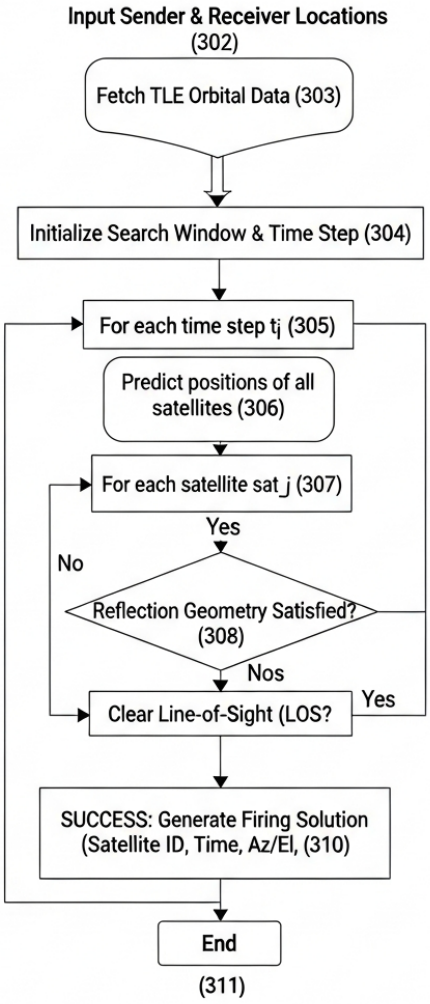

Input Sender & Receiver Locations
(302)
Fetch TLE Orbital Data (303)
Initialize Search Window & Time Step (304)
For each time step t_j (305)
Predict positions of all satellites (306)
For each satellite sat_j (307)
Yes
No
Reflection Geometry Satisfied? (308)
Nos
Yes
Clear Line-of-Sight (LOS?
SUCCESS: Generate Firing Solution (Satellite ID, Time, Az/El, (310)
End
(311)

FIG. 3

FIG. 4A Movable Antenna Mode

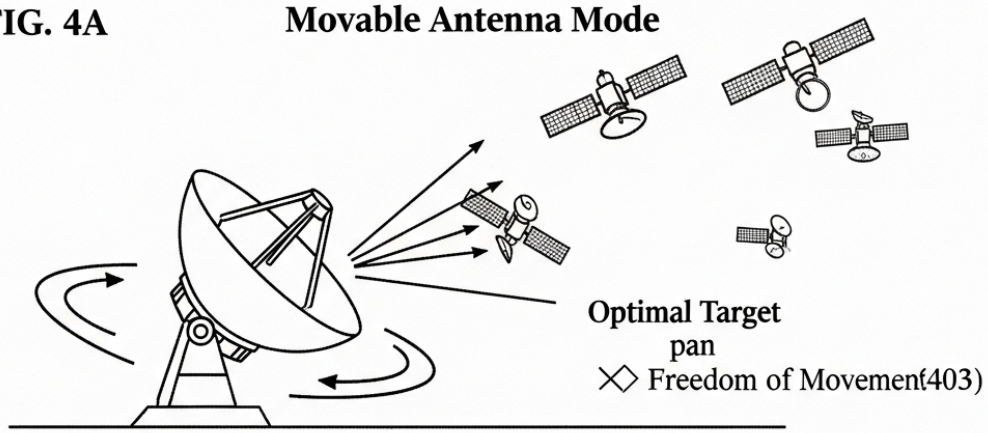


FIG. 4B Fixed Antenna Mode

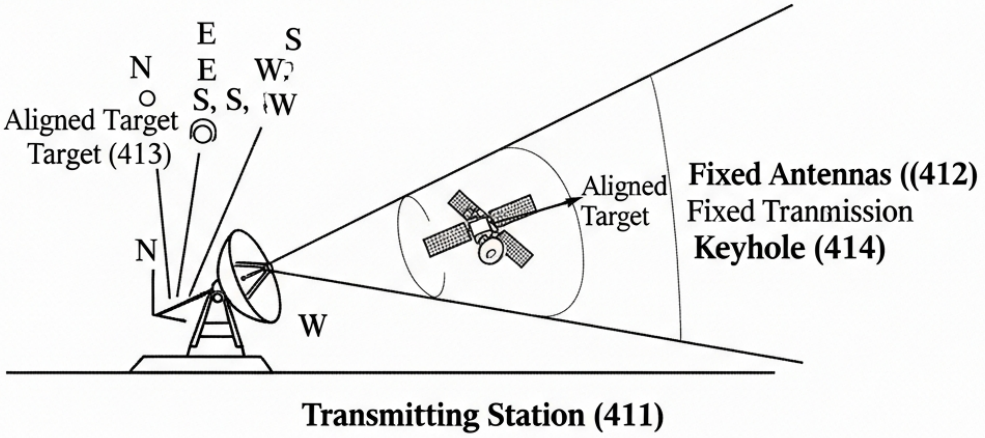


FIG. 4